

Artifact Comparison and Evaluation Web Application with Human Study Support

Group Members

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Motivation

Modern software engineering produces many different artifacts, such as source code, test cases, design diagrams, and documentation, created by both humans and automated tools such as artificial intelligence and large language models (LLMs). Evaluating the quality, correctness, understandability, and usability of these works is important but often subjective, and time-consuming. Researchers currently need an unified platform to design controlled, blinded studies and collect participant feedback.

Our project will solve this problem by providing a web-based application where researchers can upload and organize artifacts, create evaluation tasks, and gather reliable participant responses through comparison features. This platform will make artifact evaluation more systematic, objective, efficient, and easy, especially when comparing human and AI-generated outputs.

Goals

The goal of the project is to build a web application that enables researchers to upload, organize, and manage diverse software artifacts; design evaluation studies; and recruit participants who can compare and rate these artifacts under certain conditions. The system will also support participant assessment through quizzes and provide comprehensive dashboards to facilitate the use of the application for participants, researchers and reviewers.

Problem Solved

Our application fills an important gap by offering a single, flexible environment for running human-subject studies on software artifacts. Instead of using many separate tools, it combines artifact management, participant recruitment, comparison and rating in one modular web platform. This makes the researcher's job easier to plan and repeat, while keeping everything in an organized and efficient way.

Features

- **User & Role Management** – Role-based access control guarantees that users only access the functionality relevant to their role.
- **Artifact Management** – Upload, tag, search, and edit metadata for diverse artifact types (source code, test cases, diagrams, and documentation).
- **Participant Competency Assessment** – Background questionnaires and technical quizzes to match participants with studies.
- **Artifact Comparison & Annotation** – Side-by-side or 3-way artifact comparison with highlighting, inline comments, and researcher-defined rating criteria.
- **Study Management** – Create studies, assign participants, collect data, and export results securely.
- **Dashboards** – Admin dashboards to manage the website, Researcher dashboards to conduct studies and participant dashboards to track progress and assignments.
- **Deployment Options** – Local deployment using Docker or hosted on a server environment.
- **Future-Ready Architecture** – Modular and object-oriented design for easy integration of new artifact types or tools

Extra Features

- **Ranking System**
- **Notification Center**

Selling Points

- **Blinded Evaluations** reduce bias and improve research validity.
- **Flexible Architecture** easily accommodates new artifact types or metrics.
- **Practical UI** makes the system easier to use.
- **Comprehensive and User Friendly Dashboards** provide real-time insights and exportable reports.
- **AI-Assisted Quiz Generation** facilitates participant evaluation ensuring that candidates have fundamental knowledge about the research field.

Why It's Interesting

This platform makes research easier, faster, and more reliable. It not only facilitates the study process but also helps researchers quickly find and recruit the right participants from a large pool. By combining human evaluation with AI-assisted tools, it turns what is usually an unstructured and inconsistent process into a clear and repeatable workflow. Researchers can run controlled comparisons with minimal effort using features such as blinded study execution and participant skill checks through AI-generated quizzes. These tools cut down on manual work, reduce bias, and improve the quality and reproducibility of results. At the same time, participants enjoy a simple, user-friendly interface where they can complete comparison tasks and give structured feedback—leading to better data and a smoother overall experience.